



Production performance, nutrient digestibility, and enteric methane emissions of lactating Holstein cows fed triticale silage of different maturities in different dietary forage inclusions

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ABSTRACT

This study evaluated the production performance, milk fatty acid profile, nutrient digestibility, and enteric methane (CH₄) emissions of high-producing dairy cows fed high-forage (HF; 52% forage) or low-forage (LF; 37% forage) diets containing triticale silages harvested at either the boot (BT) or soft-dough (SFT) stage. The BT stage silage contained 16.7% CP, 51.1% NDF, 35.0% ADF, 3.7% ADL, and 2.2% starch, whereas the SFT stage silage contained 8.7% CP, 62.6% NDF, 46.1% ADF, 6.4% ADL, and 4.6% starch. The experiment followed a replicated 4 × 4 Latin square design with 21-d periods using 8 primiparous and 16 multiparous Holstein cows. Treatments followed a 2 × 2 factorial arrangement of forage inclusion level and triticale maturity stage. Forage maturity did not affect DMI in cows fed LF diets, but cows fed the HFBT diet consumed more DM than those fed the HFSFT diet (26.3 vs. 23.4 kg/d). Cows fed BT stage diets produced more milk than those on SFT stage diets (46.9 vs. 43.4 kg/d), whereas cows on LF diets produced more milk than those on HF diets (47.1 vs. 43.2 kg/d). The production of ECM did not differ between maturity stages but was greater for cows consuming LF diets compared with HF diets (49.1 vs. 45.6 kg/d). Cows consuming BT stage diets exhibited greater apparent total-tract digestibility of DM (69.3% vs. 67.3%) and NDF (59.4% vs. 54.5%) than SFT stage diets. Cows consuming HF diets exhibited greater digestibilities of DM (70.2% vs. 66.4%), NDF (60.3% vs. 55.8%), and starch (95.5% vs. 94.8%) than LF diets. Cows fed BT stage diets produced 23 g/d less CH₄ than cows on SFT stage

diets (368 vs. 391 g/d), with no difference between HF and LF diets in absolute CH₄ output. However, CH₄ yield was lower in cows consuming BT stage diets compared with SFT stage diets on a DMI basis (13.6 vs. 15.2 g/kg DMI). Cows consuming LF diets exhibited reduced CH₄ yields compared with HF diets (13.2 vs. 15.6 g/kg DMI). In conclusion, dairy production appears relatively resilient to differences in triticale silage maturity. However, harvesting at the boot stage offers a methane mitigation strategy by reducing CH₄ yield. Low-forage diets improved milk yield and reduced CH₄ emissions at the expense of nutrient digestibility.

Key words: small grains, cereal grain silage, small-grains silage, maturity

INTRODUCTION

Including forages in the rations of high-producing dairy cattle is essential for maintaining rumen health by supplying physically effective NDF (peNDF; Mertens, 1997) and providing energy through the fermentation of structural carbohydrates. In dairy farming systems where small-grain grasses are harvested for feed (Harper et al., 2017; Brown et al., 2018; Ferreira et al., 2025), rations for high-producing cattle may include triticale silage, or silages of other small-grain grasses, as a source of peNDF. The maturity stage at which farmers harvest these small-grain grasses is a pre-harvest decision affecting the nutritional composition and digestibility of the resulting silage (Coblentz et al., 2018a; Ferreira et al., 2025).

Deciding when to harvest small-grain grasses for silage can be controversial (Ferreira et al., 2025). The existing forage inventories, crop rotation, and nutrient management plan are some of the many factors determining harvesting time. From a nutritional perspective, Coblentz et al. (2018a,b) and Ferreira et al. (2025) reported

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

that harvesting small-grain grasses at the boot stage of maturity provides a forage with a greater concentration of CP, a lower concentration of NDF, and a higher NDF digestibility than harvesting small-grain grasses at the soft-dough stage of maturity. From a forage yield perspective, however, Coblenz et al. (2018a,b) and Ferreira et al. (2025) reported that harvesting small-grain grasses at the boot stage of maturity yields substantially less forage than harvesting small-grain grasses at the soft-dough stage of maturity. When factoring in DM yield, estimated silage costs, and forage quality, a least-cost ration formulation approach suggested that the maturity of small-grain grasses at harvest has minimal effect on overall dairy cattle feeding costs (Ferreira et al., 2025). Ferreira et al. (2025) suggested that the optimal harvesting time of triticale for silage may depend on how sensitive dairy cattle performance is to the nutritional differences between boot and soft-dough stage silages.

Schultz et al. (2025) reported no differences in DMI, ECM yield, NDF digestibility, or de novo fatty acid concentration in milk fat between cows fed diets containing boot or soft-dough stage triticale silages, suggesting that dairy cattle performance is not sensitive to triticale silage maturity. However, that study only achieved modest differences in silage nutrient composition, despite a 26-d gap between harvests. Specifically, Schultz et al. (2025) reported that triticale silage harvested in the boot stage contained 1.3 percentage units more CP and 2.0 percentage units, 2.0 percentage units, and 1.3 percentage units less NDF, ADF, and ADL, respectively (DM basis) than triticale silage harvested in the soft-dough stage, which were less than expected (Coblenz et al., 2018a; Ferreira et al., 2025). Because the study of Schultz et al. (2025) may have been limited by its harvesting methods before feeding triticale silages to dairy cattle, the sensitivity of dairy cattle performance to triticale silage maturity deserves further evaluation.

Moreover, no study to date has examined the influence of harvesting maturity of triticale for silage on enteric methane (CH₄) emissions when included in rations of lactating dairy cows. Enteric CH₄ emissions from ruminants are heavily modulated by DMI and the digestibility of DM and NDF, such that increases in DMI and digestibility are associated with increased ruminal fermentation and CH₄ production (Hristov et al., 2013; Marumo et al., 2023). Therefore, disparities in forage quality of triticale at different maturities likely influence CH₄ emissions. Therefore, research must evaluate how small-grain grass maturity at harvest influences dairy cattle performance, nutrient utilization, and CH₄ emissions.

In the present study, we hypothesized that cows fed diets containing triticale silage harvested at the boot stage of maturity would outperform cows fed diets containing triticale silage harvested at the soft-dough stage of ma-

turity due to differences in forage quality. In addition, we hypothesized that boot stage-harvested silage diets might lead to greater CH₄ production due to increased DMI, improved digestibility, and greater fermentation of structural carbohydrates. Also, we anticipated that the disparities in performance and CH₄ production would be more pronounced in diets with greater forage inclusion. Accordingly, the objective of this study was to evaluate the production performance, milk fatty acid profile, nutrient utilization, and CH₄ emissions of high-producing dairy cows consuming high- or low-forage diets containing triticale silages harvested at either the boot stage or soft-dough stage of maturity.

MATERIALS AND METHODS

Triticale Silages

Triticale was cultivated at Kentland Farm (Blacksburg, VA). A single 10-ha field was seeded on October 18, 2022, with 105 kg/ha of triticale (TriCal Grainer 154; Butte, MT). Corn for silage was harvested from the field on September 22, 2022. On November 18, 2022, 45 kg N/ha from dairy manure was applied, and on March 18, 2023, an additional 45 kg N/ha was broadcast as urea and ammonium nitrate. Starting on April 18, 2023, the growth stage of the crop was monitored weekly to assess maturity following the method outlined by Ferreira et al. (2025) and using the Zadoks scale (Zadoks et al., 1974) to determine the appropriate maturity stage for harvest.

The field was divided into 4 quadrants, and alternate quadrants were harvested for each maturity stage to reduce field effects on the nutritional qualities of the resulting crops (Schultz et al., 2025). The crop was harvested at 2 distinct stages of maturity: the boot (**BT**) stage on April 29, 2023, and the soft-dough (**SFT**) stage on May 28, 2023. At the BT stage, half of the crop was cut using a mower with conditioner rolls (John Deere 946 MoCo; Moline, IL), wilted for 2 d (to obtain $\geq 30\%$ DM concentration), tedded, and chopped with a forage chopper (John Deere 7400) set to a theoretical length of cut of 19 mm. A microbial inoculant (Early Sile Plus, Micron Bio-Systems Inc., Buena Vista, VA) was applied to the biomass during chopping at a rate of 10 mL/kg (on a fresh-weight basis). At the SFT stage, the other half of the crop was similarly cut, wilted for less than 12 h, tedded, chopped with the same theoretical length of cut, and ensiled similarly. Immediately after chopping, the biomass was ensiled in separate concrete-walled bunker silos, each lined with a plastic sheath on the floor and walls. The ensiled material was then covered with a plastic sheet and a green mesh tarp (Secure Cover, Bag Man LLC), which were secured with silo tire sides. The ensiling period lasted for a minimum

of 42 d before the first sampling and 82 d before the inclusion in dairy rations.

Multiple subsamples were collected and composited on July 10, 2023, to assess the chemical composition of each silage. The composite samples were dried in a forced-air oven at 55°C until they reached constant weight, then ground to pass through a 1 mm screen using a Wiley mill (Thomas Scientific, Swedesboro, NJ). Ground samples were sent to a commercial laboratory (Cumberland Valley Analytical Services, Waynesboro, PA) for preliminary nutritional analysis using near-infrared reflectance spectroscopy (NIR 2 Package).

Animals, Housing, and Diets

All procedures involving animals were approved by the Virginia Tech Institutional Animal Care and Use Committee. The experiment was designed as a replicated 4×4 Latin square design with 21-d periods and a 2×2 factorial arrangement of treatments, where forage maturity (BT vs. SFT) and dietary forage inclusion (high forage [HF] vs. low forage [LF]) were the main factors. The study included 16 multiparous (664 ± 53 kg BW, 111 ± 16 DIM) and 8 primiparous Holstein dairy cows (559 ± 45 kg BW, 84 ± 19 DIM). Cows were allocated to Latin squares based on parity (1, 2, or ≥ 3), milk yield, and DIM. Within each square, cows were randomly assigned to 1 of 4 diets, balancing for carry-over effects.

The HF diets were formulated to contain 52% forage, and the LF diets were formulated to contain 37% forage. A triticale silage with a nutrient profile reflecting the average of the 2 triticale silages was used to initially formulate the HF and LF diets with corn silage as a secondary forage to meet the requirements of second-parity dairy cows (605 kg BW and 82 DIM) consuming 24 kg of DM per day and producing 44 kg of milk per day containing 3.75% fat and 3.05% protein using version 3.0.8.1 of CPM Dairy (CAHP Software Information, Philadelphia, PA). Then, the individual silages (BT and SFT) replaced the average to create the final 4 rations (Table 1). Two concentrate mixtures were ordered from a commercial feed mill (Exchange Milling Company Inc., Rocky Mount, VA), which supplied all the dietary ingredients except the forages.

All cows were housed in a 24-stall pen within a freestall barn. Cows were fed once daily (1100 h) via a Calan gate system (American Calan Inc., Northwood, NH). Cows were trained to find their specific door for a 7-d period before the beginning of the study. The TMR were delivered to feed ad libitum and to obtain ~5% of feed refusals. The amounts of feed offered and feed refused were measured daily. Cows were milked twice daily (0100 h and 1200 h), and individual milk weights were automatically recorded at each milking. For statistical analyses, the average daily milk yield and DMI were

determined from d 15 to 21 of each experimental period. Average DMI included DMI from the bait pellet used for measuring enteric CH₄ emissions. The average DMI of the bait pellet was estimated from the number of cow visits to the GreenFeed system (C-Lock Inc., Rapid City, SD) and the average pellet DM deployment per drop.

Sample Collection and Analysis

Samples of feed ingredients and TMR were collected twice weekly and composited by period. A subsample of the period-composited triticale and corn silages was submitted to a commercial laboratory (Cumberland Valley Analytical Services) for fermentation profile analysis. All samples were dried in a forced-air oven at 55°C until constant weight and ground to pass through a 1-mm screen of a Wiley mill (Thomas Scientific). The concentration of ash was determined after combusting samples in a furnace (Thermolyne 30400, Barnstead International) for 3 h at 600°C (method 942.05; AOAC International, 2019). The concentration of CP was calculated as percent N $\times 6.25$ after combustion analysis of N (method 990.03; AOAC International, 2019) using a Vario El Cube CN analyzer (Elementar Americas Inc.). The concentration of ash-free NDF was determined using the Ankom200 Fiber Analyzer (Ankom Technology) with sodium sulfite and α -amylase (Ferreira and Mertens, 2007). Forage samples were additionally analyzed for ADF and ADL, which were determined sequentially. Concentrations of ADF were determined using the Ankom200 Fiber Analyzer. After determining ADF residue weights, residues were incubated for 3 h in 72% sulfuric acid within a 4-L jar that was placed in a DaisyII Incubator (Ankom Technology). Starch concentration was determined using the acetate buffer method (Hall, 2009) with α -amylase from *Bacillus licheniformis* (FAA, Ankom Technology) and amyloglucosidase from *Aspergillus niger* (E-AMGDF, Megazyme International, Wicklow, Ireland).

Milk samples (morning and afternoon milkings) were collected on d 17 and 18 of each period. Concentrations of milk fat, true protein, lactose, and MUN were analyzed by Lancaster DHIA (Manheim, PA). Weighted average component yields were calculated according to the milk yields of each sampling. Energy-corrected milk yield (kg) was estimated by deriving the equation of Tyrrell and Reid (1965) as described in Equation 1:

$$\text{ECM} = 0.327 \times MY + 12.95 \times FY + 7.65 \times PY, \quad [1]$$

where *MY* is the daily milk yield (kg), *FY* is the daily milk fat yield (kg), and *PY* is the daily protein yield (kg). Additional milk samples (morning and afternoon milkings) were collected on d 18 to determine the milk fatty acid profile. Milk fatty acids were extracted and methyl-

Table 1. Ingredient and nutritional composition of high-forage (HF) or low-forage (LF) diets containing silage of triticale harvested in the boot (BT) or soft-dough (SFT) stage of maturity¹

Item	HF		LF	
	BT	SFT	BT	SFT
Ingredient, % DM				
Corn silage	26.3	26.3	18.8	18.8
Triticale silage, BT	25.3	—	18.3	—
Triticale silage, SFT	—	25.3	—	18.3
Corn grain	19.5	19.5	23.0	23.0
Soybean meal expeller	18.0	18.0	13.5	13.5
Soybean hulls	5.25	5.25	12.3	12.3
Corn distillers grains	—	—	4.18	4.18
Whole-roasted soybeans	—	—	4.01	4.01
Supplemental fat ²	1.82	1.82	1.88	1.88
Bentonite	0.81	0.81	0.84	0.84
Calcium carbonate	0.81	0.81	0.84	0.84
Salt	0.51	0.51	0.52	0.52
Sodium bicarbonate	1.01	1.01	1.04	1.04
Magnesium oxide	0.16	0.16	0.17	0.17
Trace mineral premix ³	0.04	0.04	0.04	0.04
Vitamins A, D ₃ , E ⁴	0.04	0.04	0.05	0.05
Vitamin E, 44 IU/g	0.06	0.06	0.06	0.06
Larvicide ⁵	0.04	0.04	0.04	0.04
Ionophore ⁶	0.41	0.41	0.43	0.43
Composition, % DM				
CP	13.5 (1.1)	11.4 (1.0)	14.1 (1.1)	12.6 (1.1)
NDF	32.2 (2.1)	35.1 (1.8)	33.1 (1.7)	35.2 (1.5)
Forage NDF	23.3 (1.3)	26.2 (1.0)	16.7 (0.9)	18.9 (0.7)
Forage uNDF	4.7 (0.2)	8.7 (0.3)	3.4 (0.2)	6.2 (0.2)
Forage pdNDF	18.5 (1.4)	17.5 (1.1)	13.3 (1.0)	12.6 (0.8)
Starch	25.7 (1.2)	26.3 (1.2)	24.8 (2.0)	25.2 (2.1)

¹Values represent the mean (SD) of the estimated nutrient composition of the TMR, calculated by reconstituting individual ingredient compositions (4 composite samples by period) according to their inclusion rates.

²Energy Booster 100 (Hubbard Feeds, Mankato, MN).

³Contained 30.7% calcium; 6.3% sulfur; 35,800 mg/kg zinc; 27,400 mg/kg manganese; 6,269 mg/kg copper; 1,548 mg/kg iron; 1,190 mg/kg iodine; 524 mg/kg cobalt.

⁴Contained 24.3% calcium; 9,900 kIU/kg vitamin A; 2,200 kIU/kg vitamin D₃; 3,300 IU/g vitamin E.

⁵ClairiFly (0.67%), Central Garden & Pet Company; Schaumburg, IL.

⁶Rumensin 90 (200 g monensin/kg), Elanco Animal Health, Indianapolis, IN.

ated according to Chouinard et al. (1999) and analyzed as described by Yang et al. (2017).

Enteric Methane Emissions

Enteric CH₄ emissions were measured using a single GreenFeed unit (hereafter unit) accessible to all cows within the pen with 24 stalls. The unit was maintained and calibrated following the manufacturer's recommendations. The CO₂ recoveries averaged 103.9% ± 6.6% (n = 4). Cows were trained to the unit for a 7-d period before the beginning of the study. Methane production was estimated from voluntary visits of cows to the unit to receive a palatable bait pellet containing 17% CP, 33% NDF, and 26% starch on a DM basis. Cows visited the unit at least twice daily to ensure sufficient data collection. Visits were monitored, and cows were guided when necessary. On average, each cow visited the unit 22 ± 5 times during the 7-d data collection period. For statistical analyses,

the average daily CH₄ production was determined from d 15 to 21 of each experimental period. Average DMI, DM digested, and ECM yield from d 15 to 21 for the experimental periods were used to calculate CH₄ yield (g of CH₄/kg DMI and g of CH₄/kg DM digested) and CH₄ intensity (g CH₄/kg ECM), respectively.

Nutrient Digestibility

Apparent total-tract nutrient digestibility was estimated using lanthanum chloride (LaCl₃) as an external marker, which was sprayed on soybean hulls included in the concentrate mixture at a rate of 1 kg per cow per day, as described by Yang et al. (2017). In each period, individual fecal grab samples were collected every 6 h (skipping 2 h every 4 samples) from d 17 to 21. A total of 12 samples were collected per cow per period, dried at 55°C until constant weight, composited, ground to pass through a 1-mm screen of a Wiley mill, and stored until

analysis. The concentrations of La in the TMR and feces were determined by inductively coupled plasma optical emission spectrometry (Arcos II ICP-AES, Spectro Analytical Instruments GmbH) as described by Galyon et al. (2024). The concentrations of CP, NDF, and starch were determined as previously described.

Apparent DM digestibility (DMD; %) and nutrient digestibilities (NutrD; %) were calculated according to Equations 2 and 3, respectively:

$$\text{DMD} = 100 - \frac{\text{Dietary}[La]}{\text{Fecal}[La]} \times 100, \quad [2]$$

$$\text{NutrD} = 100 - \frac{\text{Dietary}[La]}{\text{Fecal}[La]} \times \frac{\text{Fecal}[Nutr]}{\text{Dietary}[Nutr]} \times 100, \quad [3]$$

where $[La]$ is the concentration of La (g/kg DM) in the TMR or feces and $[Nutr]$ is the concentration of either CP, NDF, or starch (% DM) in the TMR or feces. To account for the intake of the bait pellet, a weighted average TMR composition for each cow was calculated based on the nutrient profiles of the TMR by reconstituting individual ingredient nutrient compositions and the bait pellet and the average intake of each cow.

Ruminal In Situ NDF Degradability

The rates of NDF degradation of the forages were determined using an in situ ruminal degradation study according to Ferreira et al. (2022). A 0.25-g sample of each forage was placed in filter bags (F57, Ankom Technology) previously rinsed with acetone. Duplicate samples were incubated for 0, 3, 6, 12, 24, 48, 96, and 240 h in the rumen of 2 rumen-cannulated lactating dairy cows consuming a ration containing 42% corn silage, 5% triticale silage, 3% mixed-grass hay, and 50% concentrate mix, resulting in 4 replicates per sample for each time point. All bags were inserted simultaneously at feeding time (0900 h) into the rumen and extracted at the designated time points. After extraction, the bags were rinsed with tap water in three 5-min cycles using a portable washing machine (SKY2767, Best Choice Products, Irvine, CA), then dried in a forced-air oven at 55°C. The NDF residue was then weighed.

The degradation rate (K_d) of potentially degradable NDF (pdNDF) was determined using the NLIN procedure of SAS (version 9.4, SAS Institute Inc., Cary, NC) according to Equation 4:

$$\text{ISNDFD} = \text{pdNDF} \times \left\{ 1 - e^{-K_d \times (T - \text{Lag})} \right\}, \quad [4]$$

where ISNDFD is the degraded NDF (% NDF) at time T when T is greater than Lag , Lag (h) is the discrete lag time of NDF degradation (Mertens, 1973), and pdNDF is the potentially degradable NDF (% NDF) during a 240-h ruminal fermentation estimated by subtracting the remaining undegraded NDF at 240 h (uNDF) on a NDF basis from 100. The discrete lag time was estimated using simple linear regression as $\text{ISNDFU} = a + b \times \ln_{(T)}$, where ISNDFU is the remaining undegraded NDF residue at time T on a NDF basis, a represents the intercept, b is the predicted slope, and $\ln_{(T)}$ is the natural log of fermentation time. The Lag was derived according to Equation 5:

$$\text{Lag} = e^{\left[\frac{(a-100)}{-b} \right]}. \quad [5]$$

Statistical Analysis

Before designing the experiment, a statistical power analysis was performed using the POWER procedure of SAS. Considering a statistical power equal to 0.80, a probability of committing a type I error (α) equal to 0.05, and an SD equal to 2.0 kg/d for ECM, 34 experimental units per main effect was determined to be sufficient to detect a 2.0-kg/d difference in ECM production.

All variables were analyzed using the MIXED procedure of SAS. The statistical model for the variables of interest included the fixed effects of square ($df = 5$) and diet ($df = 3$) and the random effects of period ($df = 3$), cow within square ($df = 18$), and the random residual error. Orthogonal contrasts were used to test the main effects of forage maturity (BT vs. SFT), dietary forage (HF vs. LF), and their interaction. When an interaction existed, Tukey's pairwise comparisons contrasted combinations of forage maturity and dietary forage. Statistical significance was declared at $P < 0.05$, and a tendency to differ was declared at $P < 0.10$.

RESULTS AND DISCUSSION

This study aimed to determine dairy cattle performance, nutrient digestibility, and enteric CH_4 emissions when fed silage of triticale harvested at different maturities. Schultz et al. (2025) previously evaluated dairy cattle performance when fed silage of triticale harvested at either the boot or soft-dough stage and found no significant differences in DMI, ECM yield, de novo fatty acid concentrations, or NDF digestibility. However, the harvested crops and resulting silages were nutritionally similar. The approach of harvesting one side of the field at the boot stage and the other at the soft-dough stage may have undermined the intended differences in forage

Table 2. Nutritional composition and parameters of fiber digestion kinetics of forages included in the experimental diets¹

Item	Corn silage	Triticale BT	Triticale SFT
Nutrient composition ²			
CP, %	7.2 (0.8)	16.7 (1.4)	8.7 (0.8)
NDF, %	39.3 (5.4)	51.1 (1.4)	62.6 (2.0)
ADF, %	24.6 (3.0)	35.0 (0.7)	46.1 (1.7)
ADL, %	3.0 (0.4)	3.7 (0.3)	6.4 (0.3)
uNDF, %	11.4 (0.4)	6.8 (0.9)	22.4 (1.6)
uNDF, % NDF	29.4 (4.3)	13.4 (1.5)	35.8 (2.1)
pdNDF, %	27.9 (5.4)	44.2 (0.5)	40.2 (1.6)
pdNDF, %NDF	70.6 (4.3)	86.6 (1.5)	64.3 (2.1)
Starch, %	39.2 (4.8)	2.2 (0.4)	4.6 (1.0)
Degradation kinetics			
Discrete lag, h	6.3 (0.4)	3.7 (0.3)	5.0 (0.9)
Degradation rate of pdNDF, %/h	1.14 (0.14)	2.33 (0.24)	1.53 (0.23)
pH	3.82 (0.15)	4.31 (0.22)	4.05 (0.09)
Fermentation profile, % DM			
Ammonia	0.99 (0.25)	2.48 (0.44)	1.33 (0.09)
Lactic acid	4.78 (1.27)	8.10 (0.89)	6.25 (0.61)
Acetic acid	1.84 (0.72)	1.52 (0.56)	2.49 (0.69)
Propionic acid	0.00 (0.00)	0.09 (0.11)	0.07 (0.10)
Butyric acid	0.00 (0.00)	0.10 (0.20)	0.00 (0.00)
1,2-Propanediol	0.04 (0.08)	0.18 (0.13)	0.12 (0.24)
Total VFA	6.61 (1.98)	9.81 (1.59)	8.82 (1.29)

¹Values represent the mean (SD) of 4 composited samples by period per forage. BT = boot stage of maturity; SFT = soft-dough stage of maturity.

²DM basis, unless otherwise stated.

quality due to variations in growing conditions across the field (Schultz et al., 2025). This study alternatively took the approach of dividing the field seeded with triticale into 4 quadrants and harvested alternate quadrants to reduce field effects while achieving the 2 harvesting maturities. As such, we obtained substantially different forages with the same variety of triticale and in the same field as reported by Schultz et al. (2025).

Based on the composited samples over the 4 experimental periods, the silage of triticale harvested in the BT stage contained 16.7% CP, 51.1% NDF, 35.0% ADF, 3.7% ADL, and 2.2% starch, whereas the silage of triticale harvested in the SFT stage contained 8.7% CP, 62.6% NDF, 46.1% ADF, 6.4% ADL, and 4.6% starch (Table 2). As such, harvesting triticale for silage in the BT stage resulted in 8.0 percentage units more CP and 11.5 percentage units, 11.1 percentage units, and 2.7 percentage units less NDF, ADF, and ADL, respectively, than harvesting in the SFT stage. Across 2 yr, Coblenz et al. (2018a) reported that triticale harvested in the BT stage contained 8.0 percentage units more CP and 8.5 percentage units, 8.3 percentage units, and 2.4 percentage units less NDF, ADF, and ADL, respectively, relative to harvesting in the SFT stage. Similarly, Ferreira et al. (2025) found that triticale harvested in the BT stage contained 4.5 percentage units more CP and 9.1 percentage units, 8.4 percentage units, and 2.5 percentage units less NDF, ADF, and ADL, respectively, relative to harvesting in the SFT stage.

The NDF degradation kinetics parameters of the BT and SFT stage triticale silages differed substantially (Table 2). On an NDF basis, the BT stage silage contained 13.4% uNDF and 86.6% pdNDF, whereas the SFT stage silage contained 35.8% uNDF and 64.2% pdNDF. Furthermore, the BT stage silage had a shorter NDF degradation lag time (3.7 vs. 5.0 h) and a faster K_d of the pdNDF (2.33 vs. 1.53%/h) than the SFT stage silage. Whereas K_d reported by Coblenz et al. (2018b) were overall faster (5.90%/h for BT vs. 3.10%/h for SFT), these differences may be attributed to variations in triticale varieties or estimation methods. Collectively, harvesting triticale for silage in the BT stage of maturity yields a higher-quality crop with greater fiber digestibility relative to the SFT stage of maturity.

Two primary approaches are commonly used to formulate experimental diets for forage evaluation. The first approach aligns diets nutritionally by accounting for the nutritional composition of the forages (Schultz et al., 2025). Although this method mirrors field conditions, it often results in diets with significantly different ingredient compositions. For example, in our study, the silage of triticale harvested at the SFT stage contained less CP and more NDF compared with its BT stage counterpart. Consequently, if diets were formulated based on the nutrient composition of the silages within each forage level, the SFT stage diets would necessitate more corn silage and less triticale silage (on a DM basis) to balance for differences in CP and NDF concentrations. This approach

Table 3. Production performance of lactating dairy cows consuming high-forage (HF) or low-forage (LF) diets containing silage of triticale harvested in the boot (BT) or soft-dough (SFT) stage of maturity

Item	HF		LF		SEM	P-value ¹		
	BT	SFT	BT	SFT		M	F	M × F
Milk yield, kg/d	44.9	41.4	48.9	45.3	1.2	0.01	0.01	0.98
Milk fat, %	3.81	4.19	3.60	4.07	0.17	0.01	0.20	0.73
Milk protein, %	2.85	2.81	2.92	2.88	0.06	0.06	0.01	0.97
Milk lactose, %	4.82	4.80	4.83	4.82	0.02	0.26	0.30	0.42
Milk fat yield, kg/d	1.69	1.73	1.75	1.83	0.08	0.27	0.12	0.68
Milk protein yield, kg/d	1.28	1.16	1.43	1.31	0.04	0.01	0.01	0.96
Milk lactose yield, kg/d	2.17	1.98	2.37	2.19	0.07	0.01	0.01	0.96
ECM yield, kg/d	46.4	44.7	49.6	48.5	1.4	0.09	0.01	0.74
Feed efficiency, kg ECM/kg DMI	1.82 ^{ab}	1.93 ^a	1.78 ^{ab}	1.71 ^b	0.11	0.63	0.01	0.04
MUN, mg/dL	14.4	14.0	13.8	14.7	0.9	0.51	0.82	0.08

^{a,b}Means with different superscripts within a row are different ($P < 0.05$).

¹M = effect of forage maturity (BT vs. SFT); F = effect of forage inclusion level (HF vs. LF); M × F = interaction between forage maturity and forage inclusion level.

could also necessitate additional or alternative ingredients in the concentrate of the SFT stage diets to balance the nutrient disparities and achieve a similar TMR nutrient composition as the BT stage diets. Such discrepancies can confound experimental outcomes by altering both the triticale-to-corn silage ratio and ingredient composition. The second approach involves replacing experimental forages on a one-to-one ratio (DM basis). Whereas this ensures a consistent triticale-to-corn silage ratio within forage level, this method results in variations in the diet nutrient composition when the experimental forages substantially vary in their nutrient composition. To reduce confounding effects related to the inclusion of alternative ingredients or differing triticale-to-corn silage ratios within forage levels, we adopted the second formulation method. As a result, the diets containing BT stage triticale silage had slightly more CP and less NDF than those with SFT stage triticale silage (Table 1), reflecting the inherent differences in the nutritional composition of the individual silages (Table 2). Specifically, based on the reconstituted ingredients, the diets containing BT stage triticale silage contained 1.8 percentage units more CP, 2.5 percentage units less NDF, and 0.5% unit less starch than the diets containing SFT stage triticale silage.

Cows consuming diets containing BT stage triticale produced 3.5 kg/d more milk than cows consuming SFT stage diets (46.9 vs. 43.4 kg/d; $P < 0.01$; Table 3), whereas cows consuming LF diets produced 3.9 kg/d more milk than cows consuming HF diets (47.1 vs. 43.2 kg/d; $P < 0.01$). Whereas milk fat concentration did not differ between cows consuming HF and LF diets ($P = 0.20$), cows consuming diets containing SFT stage triticale had greater milk fat concentration than cows consuming diets containing BT stage triticale (4.13% vs. 3.71%; $P < 0.01$). Additionally, milk fat yields did not differ across dietary treatments and averaged 1.75

kg/d. Cows consuming BT stage diets tended to produce milk containing a slightly greater concentration of protein compared with cows consuming SFT stage diets, although this difference was marginal (2.89% vs. 2.85%; $P = 0.06$). Cows consuming LF diets produced milk containing a greater concentration of milk protein than cows consuming HF diets (2.90% vs. 2.83%; $P = 0.01$). This may be associated with the increased amount of CP digested by cows consuming diets containing BT stage silage compared with cows consuming SFT stage silage (2.3 vs. 1.7 kg/d; $P < 0.01$; Table 4) and by cows consuming LF diets compared with HF diets (2.1 vs. 1.9 kg/d; $P < 0.01$). Cows consuming diets containing BT triticale silage had a greater milk protein yield than cows consuming diets containing SFT triticale (1.36 vs. 1.24 kg/d; $P < 0.01$). Similarly, cows consuming LF diets had a greater milk protein yield compared with cows consuming HF diets (1.37 vs. 1.22 kg/d; $P < 0.01$). Comprehensively, cows consuming BT stage diets tended to produce more ECM than cows consuming SFT stage diets (48.0 vs. 46.6 kg/d; $P = 0.09$), and cows consuming LF diets produced 3.5 kg/d more ECM than cows consuming HF diets (49.1 vs. 45.6 kg/d; $P = 0.01$).

Despite differences in DM yield, estimated silage costs, and forage quality, Ferreira et al. (2025) determined that the harvesting maturity of triticale silage has minimal effect on feeding costs for dairy cattle. From this ration formulation approach, one may determine that triticale silage can be fed to lactating dairy cattle indiscriminately regardless of maturity. However, the production response of lactating dairy cattle to differences in the nutritional composition of the resulting silages must be evaluated if they are sensitive to those differences. To evaluate this, Schultz et al. (2025) evaluated dairy cattle production response to triticale silage harvested at either the BT or SFT stages of maturity at low- and high-dietary forage in-

Table 4. Dry matter intake, nutrient intake, and nutrient digestion of lactating dairy cows consuming high-forage (HF) or low-forage (LF) diets containing silage of triticale harvested in the boot (BT) or soft-dough (SFT) stage of maturity

Item	HF		LF		SEM	P-value ¹		
	BT	SFT	BT	SFT		M	F	M × F
Intake, kg/d								
DM	26.3 ^b	23.4 ^c	28.2 ^a	28.7 ^a	1.45	0.01	0.01	0.01
CP	3.4	2.8	3.9	3.4	0.28	0.01	0.01	0.85
NDF	10.3 ^b	9.6 ^b	11.1 ^a	11.3 ^a	0.59	0.24	0.01	0.02
Forage NDF	5.4	5.1	5.6	6.0	0.74	0.90	0.01	0.08
Forage uNDF	1.4	1.5	1.5	1.8	0.30	0.04	0.09	0.36
Forage pdNDF	4.0 ^a	3.6 ^b	4.2 ^a	4.3 ^a	0.51	0.13	0.01	0.03
Starch	6.5 ^{bc}	6.1 ^c	6.9 ^b	7.5 ^a	0.36	0.49	0.01	0.01
Digestibility, %								
DM	71.1	69.2	67.5	65.3	1.28	0.03	0.01	0.92
CP	60.5 ^a	57.2 ^a	60.3 ^a	51.3 ^b	2.74	0.01	0.04	0.04
NDF	62.4	58.1	56.3	50.8	2.57	0.01	0.01	0.70
Starch	95.2	95.7	94.7	94.9	0.81	0.09	0.01	0.49
Digested, kg/d								
DM	18.9 ^a	16.3 ^b	19.1 ^a	18.9 ^a	1.00	0.01	0.01	0.01
CP	2.1	1.6	2.4	1.8	0.19	0.01	0.01	0.52
NDF	6.5	5.6	6.2	5.8	0.40	0.01	0.85	0.28
Starch	6.3 ^{bc}	5.9 ^c	6.6 ^b	7.2 ^a	0.33	0.44	0.01	0.01

^{a-c}Means with different superscripts within a row are different ($P < 0.05$).

¹M = effect of forage maturity (BT vs. SFT); F = effect of forage inclusion level (HF vs. LF); M × F = interaction between forage maturity and forage inclusion level.

clusion levels. Whereas they did not observe a significant effect of triticale silage maturity on DMI or ECM yield, Schultz et al. (2025) did not achieve substantially different forages by harvesting at the BT and SFT stage of maturity, and therefore their conclusions were limited. We repeated the study with slight modifications to the harvesting of the field to achieve substantially different forages as observed by Ferreira et al. (2025) and Coblenz et al. (2018a,b). Under the present experimental conditions, we similarly observed that feed efficiency (1.81 kg ECM/kg DMI; $P = 0.63$; Table 3) and ECM production were not affected by feeding triticale silage of different maturities.

The present study and those of Ferreira et al. (2025) and Schultz et al. (2025) suggest that triticale harvest maturity for silage may not be critical determinants of ration costs and dairy cattle production performance. However, the present study did not evaluate long-term implications, such as forage inventory and dairy cattle reproduction, body conditioning, and overall health. Thus, whereas the present findings here and in the literature (Ferreira et al., 2025; Schultz et al., 2025) indicate that triticale silage maturity may not critically affect short-term ration cost or milk production, broader conclusions should be made cautiously and within the context of diet formulation and herd management goals.

Although forage maturity did not affect the DMI of cows consuming LF diets (28.5 kg/d; $P = 0.76$), cows consuming the HF diet with BT stage triticale consumed 2.9 kg/d more DM than those consuming the HF diet with SFT stage triticale (26.3 vs. 23.4 kg/d; $P = 0.01$; Table 4). We

hypothesized that DMI would be affected by forage maturity regardless of forage inclusion level. However, in LF diets, triticale silage only constituted 18% of the dietary DM, likely attenuating its effect on rumen turnover and intake, despite still affecting overall dietary digestibility. Cows consuming diets containing BT triticale silage had greater apparent total-tract digestibilities of DM (69.3% vs. 67.3%; $P < 0.01$) and NDF (59.4% vs. 54.5%; $P < 0.01$) than cows consuming diets containing SFT triticale silage, regardless of forage inclusion level (Table 4). These differences in digestibility likely reflect inherent differences in the ruminal degradability of the 2 silages, particularly in the concentration of pdNDF and its K_d .

To better understand why DMI differed in HF but not in LF diets, it is important to consider the magnitude of differences in forage fiber fractions. Within HF diets, the BT diet contained 2.9 percentage units less forage NDF, 4.0 percentage units less forage uNDF, and 1.0 percentage units more forage pdNDF than the SFT diet (Table 1). Within LF diets, the BT diet contained 2.2 percentage units less forage NDF, 2.8 percentage units less forage uNDF, and 0.7 percentage units more forage pdNDF than the SFT diet. These narrower differences between maturities in the LF diets may have limited the extent to which ruminal fiber degradation could influence rumen fill or passage rate. Thus, whereas differences in the ruminal DM and NDF digestibilities of the 2 triticale maturities may have been sufficient to improve the total-tract digestibilities of DM and NDF of the BT diets compared with SFT diets, regardless of forage inclusion level, they

were likely not pronounced enough to elicit measurable changes in rumen kinetics, rumen fill, or DMI within LF diets (Ellis et al., 1994).

Cows consuming HF diets exhibited greater total-tract digestibilities of DM (70.2% vs. 66.4%; $P = 0.03$), NDF (60.3% vs. 55.8%; $P < 0.01$), and starch (95.5% vs. 94.8%; $P < 0.01$) than their LF counterparts. An interaction between forage maturity and forage inclusion level existed for the total-tract CP digestibility ($P = 0.04$), which was driven by the reduced CP digestibility of the LFSFT diet compared with the other 3 diets (51.3% vs. 59.3%). We anticipated that total-tract digestibility would be improved in LF diets due to the reduced forage-to-concentrate ratio and increase in fermentable carbohydrates. To balance for the reduced forage NDF in the LF diets compared with HF diets (17.8% vs. 24.8%), other fibrous ingredients in the concentrate, primarily soybean hulls, were included in greater proportions than in the HF diets (Table 1). Low-forage diets likely had a reduced mean particle size and less physically effective fiber. As a result, the ruminal passage rate could have been faster in LF diets compared with HF diets. A faster ruminal rate of passage could decrease the effective ruminal degradation of LF diets and explain the observed decreases in the apparent total-tract digestibilities of DM, CP, NDF, and starch. Likewise, faster ruminal passage rates and reduced rumen fill of LF diets could explain the increased DMI ($P < 0.01$) of cows consuming these diets relative to HF diets due to faster ruminal emptying and greater turnover.

Feed intake explains 60% to 80% of the variation in enteric CH_4 production (NASEM, 2021). However, despite the greater DM ingestion and digestion, cows consuming BT stage triticale silage produced 23 g CH_4/d less than those consuming SFT stage silage (368 vs. 391 g CH_4/d ; $P = 0.01$; Figure 1). Curiously, CH_4 production did not differ between cows consuming HF and LF diets ($P = 0.28$) even though the cows consuming LF diets ingested and digested significantly more DM. On a DMI basis, cows consuming diets containing BT stage triticale yielded less CH_4 than cows consuming diets containing SFT stage triticale (13.6 vs. 15.2 g CH_4/kg DMI; $P = 0.01$), and cows consuming LF diets yielded less CH_4 than cows consuming HF diets (13.2 vs. 15.6 g CH_4/kg DMI; $P = 0.01$). Similarly, on a DM digested basis, cows consuming diets containing BT stage triticale yielded less CH_4 than cows consuming diets containing SFT stage triticale (19.3 vs. 22.3 g CH_4/kg DM digested; $P = 0.01$), and cows consuming LF diets yielded less CH_4 than cows consuming HF diets (19.7 vs. 21.9 g CH_4/kg DM digested; $P = 0.01$).

There is a clear drive in the literature for using additives such as macroalgae and 3-nitrooxypropanol in rations of dairy cattle to reduce enteric CH_4 emissions,

especially in the recent special issue on feed additives for methane mitigation published by the *Journal of Dairy Science* in January of 2025. Whereas feed additives can significantly reduce enteric CH_4 emissions by 30% to 50% (Purba and Sangsawad, 2025), DMI is often reduced along with milk production (Fouts et al., 2022). Here, we uniquely show that reduced enteric CH_4 production can be achieved while sustaining DMI and increasing milk production by feeding triticale silage of an earlier maturity compared with a later maturity. Achieving similar DMI and greater milk yield while reducing enteric CH_4 emissions makes forage maturity a more attractive CH_4 mitigation strategy for producers to implement than paying for costly feed additives that provide no economic return (Hristov et al., 2022; Martins et al., 2025). Understanding the mechanisms behind how forage maturity influences enteric CH_4 emissions will allow producers to make informed decisions on optimal harvesting maturity for their crops while being environmentally conscientious. However, the paucity in the literature regarding the influence of forage maturity on enteric CH_4 emissions presently makes conclusions difficult.

The differences in CH_4 production and CH_4 yields between forage maturity and dietary forage inclusion level suggest alterations in ruminal fermentation, either by hydrogen dynamics or digestion kinetics within the rumen. Adams et al. (1987) evaluated the effects of advancing forage maturity of mixed-grass pasture on ruminal fermentation of cannulated steers over 5 mo and determined that advanced maturity increases the molar proportion of acetate in rumen fluid, whereas propionate molar proportion decreases and butyrate molar proportion stays relatively constant. We additionally observed that the BT stage triticale silage contained less acetate on a DM basis than the SFT stage triticale silage (Table 2). Diets with higher forage-to-concentrate ratios similarly demonstrate greater ruminal acetate-to-propionate ratios (Penner et al., 2009; Shi et al., 2018). These ruminal fermentation responses are likely linked to increased microbial cellulolytic activity due to the increased presence of cellulose and hemicellulose (Van Soest, 1994). Additionally, in the case of LF versus HF diets, LF diets may lead to proliferation of propionate-producing microbes due to the increased presence of nonforage fiber carbohydrates (Bauman et al., 1971; Wang et al., 2018) associated with the greater starch intake (7.2 vs. 6.3 kg/d; $P < 0.01$) and digestion (6.9 vs. 6.1 kg/d; $P < 0.01$) that occurred. Thus, we might expect that cows consuming the LF diets and those consuming diets containing BT stage triticale silage exhibited reduced acetate-to-propionate ratios. This potential shift in VFA proportions lends itself to decreased CH_4 emissions as less hydrogen is produced with propionate than acetate (Van Soest, 1994), thereby decreasing available hydro-

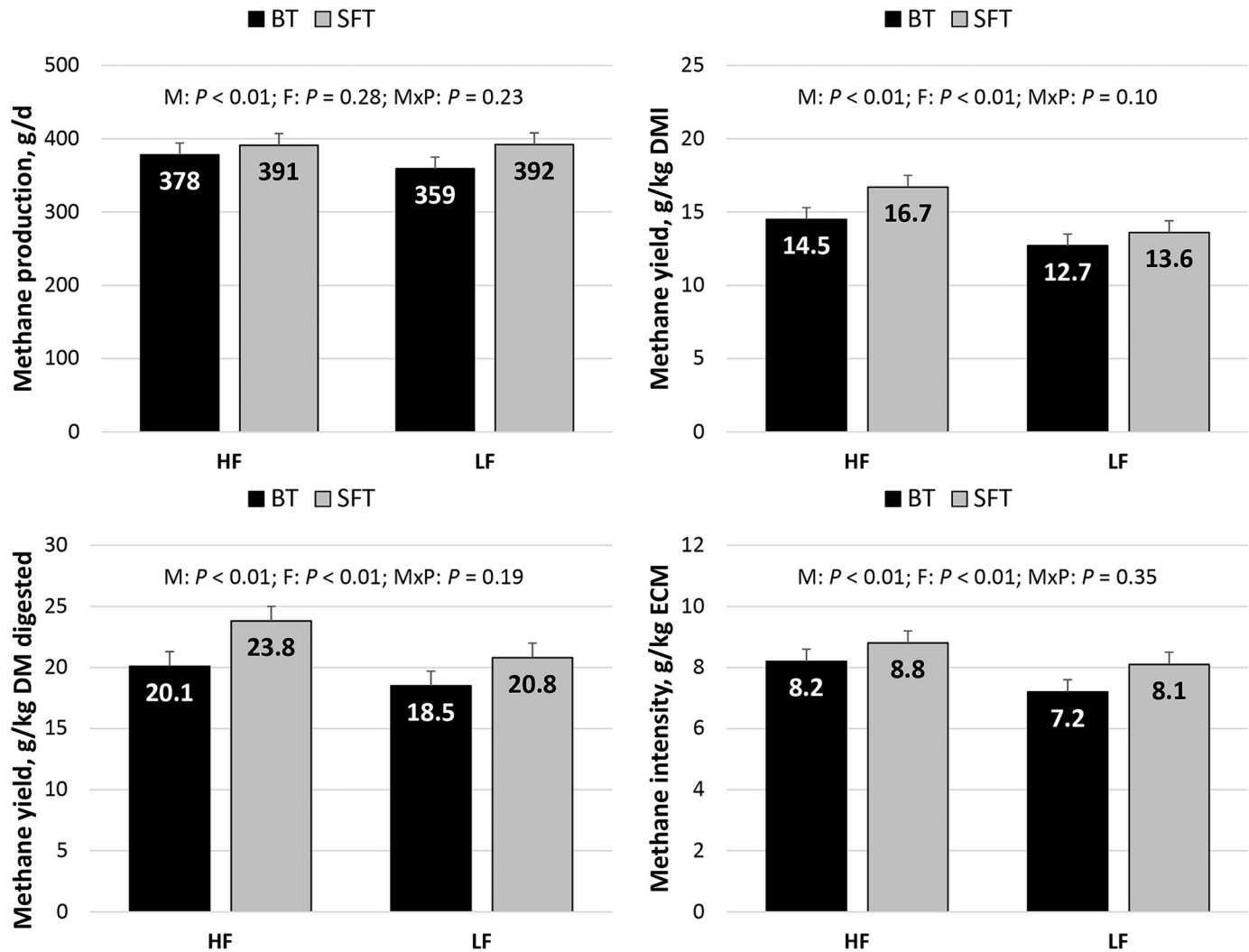


Figure 1. Methane production (g/d), yield (g/kg DMI and g/kg DM digested), and intensity (g/kg ECM) of lactating dairy cows consuming high-forage (HF) and low-forage (LF) diets containing triticale silage harvested either at the boot (BT) or soft-dough (SFT) stage of maturity. M = effect of forage maturity (BT vs. SFT); F = effect of dietary forage inclusion (HF vs. LF); M × F = interaction between forage maturity and dietary forage inclusion. Error bars represent SEM.

gen for the synthesis of CH_4 (NASEM, 2021). However, differences in molar percentages do not necessarily reflect the pools of acetate, propionate, and butyrate in the rumen. If ruminal fermentation activity was increased in cows consuming LF diets and those containing BT stage triticale, the relative pools of acetate may have been greater compared with cows consuming HF diets and those containing SFT stage triticale silage.

Regarding ruminal digestion kinetics, high-quality forages with faster K_d are often associated with faster passage rates (Ellis et al., 1994; Galyon et al., 2024). Faster passage rates could reduce ruminal fermentation of carbohydrates, which could reduce CH_4 emissions (Beauchemin et al., 2022; Hristov et al., 2022). However, if the NDF passage rates of the BT stage diets were faster

than the SFT stage diets, the balance between the NDF degradation and passage rates was not so deficient that NDF digestibility was impaired in cows consuming BT stage diets. If we assume that hindgut digestion of NDF is less than 10% of total NDF digestion (Huhtanen et al., 2006; Lopes et al., 2015), then ruminal fermentation of NDF was greater in cows consuming BT triticale silage than SFT triticale silage based on the total-tract digestibility results. Perhaps ruminal passage rate plays an alternative role in enteric CH_4 reductions than reduced ruminal fermentation. Conversely, faster ruminal passage rates in cows fed the LF diets could explain the reduced enteric CH_4 production via reduced DM and NDF digestibility relative to cows fed HF diets. This may additionally explain the reduced feed efficiency of cows

Table 5. Milk fatty acid profile (g/100 g fatty acids) of lactating dairy cows consuming high-forage (HF) or low-forage (LF) diets containing silage of triticale harvested in the boot (BT) or soft-dough (SFT) stage of maturity

Fatty acid	HF		LF		SEM	P-value ¹		
	BT	SFT	BT	SFT		M	F	M × F
4:0	3.87	3.90	3.61	3.77	0.11	0.31	0.05	0.52
6:0	2.24	2.15	2.11	2.15	0.08	0.72	0.25	0.31
8:0	1.26	1.15	1.22	1.20	0.05	0.11	0.90	0.32
10:0	2.69	2.34	2.69	2.53	0.14	<0.01	0.32	0.29
11:0	0.06	0.04	0.07	0.05	0.01	<0.01	<0.01	0.78
12:0	2.96	2.51	2.97	2.73	0.16	<0.01	0.28	0.36
13:0	0.09	0.07	0.11	0.09	0.01	<0.01	<0.01	0.77
14:0	10.06	9.10	9.71	9.20	0.33	<0.01	0.63	0.39
<i>cis</i> -9 14:1	1.07	0.92	1.06	0.89	0.07	<0.01	0.71	0.71
15:0	0.82	0.66	0.86	0.72	0.03	<0.01	0.10	0.83
<i>iso</i> 15:0	0.18	0.18	0.16	0.17	0.01	0.33	0.03	0.11
<i>anteiso</i> 15:0	0.35	0.33	0.35	0.34	0.01	0.28	0.87	0.31
16:0	30.45	29.57	27.33	27.83	0.86	0.77	<0.01	0.27
<i>iso</i> 16:0	0.15	0.16	0.14	0.16	0.01	0.09	0.46	0.64
<i>cis</i> -9 16:1	1.25	1.25	1.16	1.07	0.06	0.24	<0.01	0.19
17:0	0.41	0.39	0.39	0.39	0.02	0.39	0.45	0.59
<i>iso</i> 17:0	0.34	0.34	0.32	0.33	0.02	0.96	0.25	0.53
18:0	10.07	10.83	9.83	11.29	0.41	<0.01	0.76	0.32
<i>trans</i> -6,8 18:1	0.35	0.30	0.45	0.35	0.03	<0.01	<0.01	0.31
<i>cis</i> -9 18:1	22.15	23.34	22.03	23.14	0.72	0.11	0.82	0.96
<i>trans</i> -9 18:1	0.34	0.30	0.41	0.33	0.02	<0.01	<0.01	0.28
<i>trans</i> -10 18:1	1.28	0.81	1.94	1.20	0.23	<0.01	<0.01	0.33
<i>cis</i> -11 18:1	0.84	0.77	0.93	0.80	0.04	<0.01	0.05	0.29
<i>trans</i> -11 18:1	1.10	1.10	0.96	1.00	0.06	0.69	0.02	0.65
<i>trans</i> -12 18:1	0.49	0.40	0.54	0.47	0.03	<0.01	<0.01	0.67
<i>cis</i> -9, <i>trans</i> -11 CLA	0.50	0.48	0.45	0.44	0.03	0.27	0.03	0.75
<i>trans</i> -10, <i>cis</i> -12 CLA	0.008	0.004	0.013	0.007	0.002	<0.01	0.01	0.45
<i>cis</i> -9, <i>cis</i> -12 18:2	3.04	2.89	3.98	3.67	0.13	0.06	<0.01	0.52
<i>cis</i> -9, <i>cis</i> -12, <i>cis</i> -15 18:3	0.43	0.38	0.50	0.43	0.02	<0.01	<0.01	0.49
19:0	0.18	0.10	0.21	0.12	0.02	<0.01	0.05	0.69
20:0	0.15	0.15	0.14	0.16	0.01	0.07	0.88	0.12
Other	0.95	0.82	0.98	0.84	0.04	<0.01	0.44	0.91
De novo fatty acids ²	41.57	38.87	39.26	38.39	1.17	0.07	0.15	0.34
Preformed fatty acids ³	57.54	58.02	57.34	58.67	1.45	0.50	0.87	0.76

¹M = effect of forage maturity (BT vs. SFT); F = effect of dietary forage inclusion (HF vs. LF); M × F = interaction between forage maturity and dietary forage inclusion.

²Sum of <C16 + 0.5 × C16.

³Sum of >C16 + 0.5 × C16.

consuming LF diets compared with HF diets (1.75 vs. 1.88 kg ECM/kg DMI; $P < 0.01$; Table 4). Discussion at this point is theoretical without simultaneous measurements of the ruminal VFA profile and ruminal passage rate. Studies that specifically evaluate the relationships between forage maturity, ruminal passage rate, ruminal VFA production, and enteric CH₄ emissions are needed to disseminate these findings.

Whereas the milk fat yield of cows in this study was not affected by forage maturity or inclusion level, variations were observed in the fatty acid profile of milk from cows consuming the different diets (Table 5). Cows consuming diets containing BT stage triticale silage produced milk fat with greater concentrations of *trans*-10 18:1 (1.66 vs. 1.02 g/100 g fatty acids; $P < 0.01$) and *trans*-10,*cis*-12 CLA (0.011 vs. 0.006 g/100 g fatty acids; $P < 0.01$) compared with cows consuming SFT stage triticale silage.

Incremental concentrations of *trans*-10 18:1, *trans*-10,*cis*-12 CLA, or both in milk fatty acids are typically associated with lower, not higher, concentrations of de novo fatty acids (Griinari et al., 1998; Baumgard et al., 2001). In this study, cows consuming BT stage triticale silage tended to produce more de novo fatty acids (40.4 vs. 38.6 g/100 g fatty acids), which contradicts the expected negative relationship between *trans*-10 18:1 or *trans*-10,*cis*-12 CLA and de novo fatty acids. Nonetheless, the daily amounts of de novo fatty acids secreted in milk did not differ between cows consuming BT and SFT triticale silages ($P = 0.66$; Figure 2).

Because milk fat depression implies a substantial reduction in milk fat production (Davis, 1967; Davis and Brown, 1970; Bauman and Davis, 1974; Bauman and Griinari, 2003) due to a decrease in de novo fatty acid synthesis, we conclude that the lower milk fat concen-

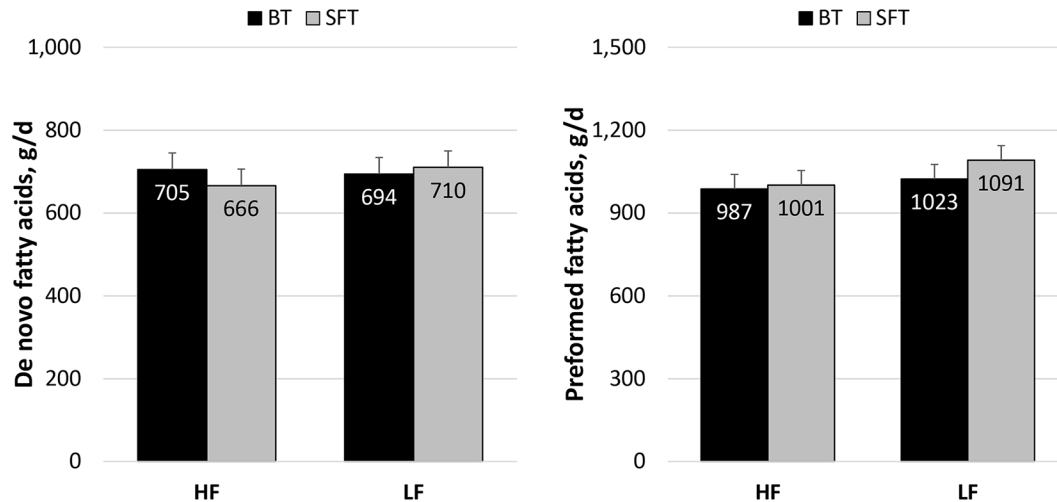


Figure 2. De novo and preformed fatty acid production (g/d) of lactating dairy cows consuming high-forage (HF) and low-forage (LF) diets containing triticale silage harvested either at the boot (BT) or soft-dough (SFT) stage of maturity. M = effect of forage maturity (BT vs. SFT); F = effect of dietary forage inclusion (HF vs. LF); M × F = interaction between forage maturity and dietary forage inclusion. Error bars represent SEM.

tration from cows consuming BT stage triticale silage is not related to milk fat depression. The reasons for the simultaneous increases in *trans*-10 18:1, *trans*-10,*cis*-12 CLA, and de novo fatty acids in the fatty acid profile remain unclear. Milk from cows consuming SFT stage triticale silage tended to have slightly greater concentrations of oleic acid in their fatty acid profile than cows consuming BT stage triticale silage (23.2 vs. 22.1 g/100 g fatty acids; $P = 0.10$). Although speculative, the oleic acid from the grain germ in triticale harvested at the soft-dough stage of maturity can plausibly explain the greater proportion of *trans*-10 18:1, *trans*-10,*cis*-12 CLA, and de novo fatty acids in the milk fatty acid profile.

One potential limitation of this study was the unexpectedly low CP concentrations in the formulated diets, which were originally intended to reflect higher CP levels. When analyzed values did not meet expectations, our first step was to reanalyze the TMR samples to verify the results. After confirming the lower-than-expected CP concentrations, we analyzed individual feed ingredients (i.e., corn silage, triticale silages, and concentrate mixtures) for each diet. This revealed that the concentrates supplied by the commercial feed mill contained substantially less CP than anticipated. To ensure analytical accuracy, we verified our nitrogen standards and confirmed the reliability of our methods, then submitted samples to an independent commercial laboratory (Cumberland Valley Analytical Services), whose results corroborated ours. We subsequently contacted the feed mill and reviewed their production logs, which confirmed that the formulas matched our original specifications. Based on these multiple layers of verification, we are confident

that the reported CP values accurately represent what was fed and analyzed, despite being lower than intended.

CONCLUSIONS

We initially hypothesized that cows consuming BT stage triticale silage would produce more enteric CH₄ due to the improved digestibility of the diet and greater DMI. Contrary to our hypothesis, cows consuming BT stage triticale produced less enteric CH₄, likely due to shifts in ruminal fermentation and passage kinetics. Feeding LF diets with these forages seems to improve dairy cow performance and reduce CH₄ emissions at the expense of digestive efficiency. From an environmental standpoint, feeding triticale silage of the BT stage maturity seems to be a sustainable CH₄ mitigation strategy that simultaneously increases DMI and milk yield. However, the mechanism behind the reduction in enteric CH₄ in response to earlier forage maturity is still speculative.

NOTES

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Nonstandard abbreviations used: BT = boot; HF = high forage; K_d = degradation rate; LF = low forage; pdNDF = potentially degradable NDF; peNDF = physically effective NDF; SFT = soft-dough; uNDF = undegraded NDF.

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